

Degeneracy and Cycling

• Degeneracy

We know that in a basic feasible solution, all nonbasic variables are equal to zero

If some basic variables are also zero, then this basic feasible solution is said to be degenerate

e.g.: $\max \quad z = 4x_1 + 3x_2$

s.t. $2x_1 + x_2 \leq 8$

$x_1 + x_2 \leq 5$

$x_1 - x_2 \leq 4$

$x_1 \leq 4$

$x_1, x_2 \geq 0$

standard form

$\max \quad z = 4x_1 + 3x_2$

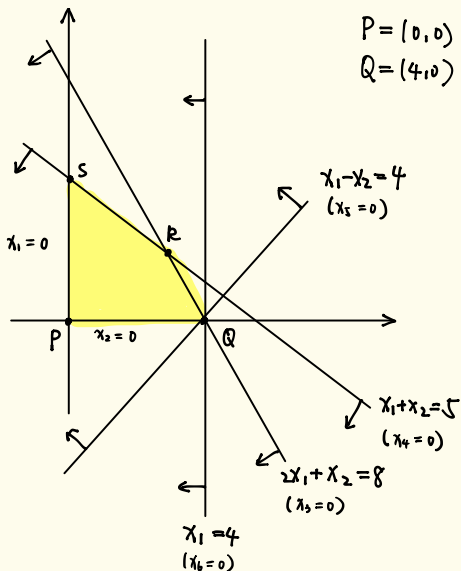
s.t. $2x_1 + x_2 + x_3 = 8$

$x_1 + x_2 + x_4 = 5$

$x_1 - x_2 + x_5 = 4$

$x_1 + x_6 = 4$

$x_1, \dots, x_6 \geq 0$



	Z	x ₁	x ₂	x ₃	x ₄	x ₅	x ₆	RHS
Z	1	-4	-3	0	0	0	0	0
x ₃	0	2	1	1	0	0	0	8
x ₄	0	1	1	0	1	0	0	5
x ₅	0	①	-1	0	0	1	0	4
x ₆	0	1	0	0	0	0	1	4

P

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	Z	x ₁	x ₂	x ₃	x ₄	x ₅	x ₆	RHS
Z	1	0	-7	0	0	4	0	16
x ₃	0	0	3	1	0	-2	0	0
x ₄	0	0	2	0	1	-1	0	1
x ₁	0	1	-1	0	0	1	0	4
x ₆	0	0	①	0	0	-1	1	0

Q

→

	Z	x ₁	x ₂	x ₃	x ₄	x ₅	x ₆	RHS
Z	1	0	0	0	0	-3	7	16
x ₃	0	0	0	1	0	①	-3	0
x ₄	0	0	0	0	1	1	-2	1
x ₁	0	1	0	0	0	0	1	4
x ₂	0	0	1	0	0	-1	1	0

Q

→

	Z	x ₁	x ₂	x ₃	x ₄	x ₅	x ₆	RHS
Z	1	0	0	3	0	0	-2	16
x ₅	0	0	0	1	0	1	-3	0
x ₄	0	0	0	-1	1	0	①	1
x ₁	0	1	0	0	0	0	1	4
x ₂	0	0	1	1	0	0	-2	0

Q

→

	Z	x ₁	x ₂	x ₃	x ₄	x ₅	x ₆	RHS
Z	1	0	0	1	2	0	0	18
x ₅	0	0	0	-2	3	1	0	3
x ₆	0	0	0	-1	1	0	1	1
x ₁	0	1	0	1	-1	0	0	3
x ₂	0	0	1	-1	2	0	0	2

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$x = (0, 1, 4, 0, 0, 0)$ is a degenerate BFS, which corresponds to Q on the graph. We observe that:

- A degenerate BFS corresponds to more than one basis matrices.
- "Too many" defining hyperplanes intersect at a degenerate extreme point.
- Degeneracy may cause cycling.

A Cycling Example

Consider the following linear program.

$$\begin{aligned}
 \max \quad & z = 10x_1 - 57x_2 - 9x_3 - 24x_4 \\
 \text{S.t.} \quad & 0.5x_1 - 5.5x_2 - 2.5x_3 + 9x_4 \leq 0 \\
 & 0.5x_1 - 1.5x_2 - 0.5x_3 + x_4 \leq 0 \\
 & x_1 \leq 1 \\
 & x_1, x_2, x_3, x_4 \geq 0
 \end{aligned}$$

We apply the simplex method with the following rule.

- Choose entering variable with most negative reduced cost.
- Choose leaving variable with smallest subscript if there is a tie in the minimum ratio test.

The simplex method works as follows:

	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	1	-10	57	9	24	0	0	0	0
x_5	0	0.5	-5.5	-2.5	9	1	0	0	0
x_6	0	0.5	-1.5	-0.5	1	0	1	0	0
x_7	0	1	0	0	0	0	0	1	1

↓

	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	1	0	-53	-41	204	20	0	0	0
x_1	0	1	-11	-5	18	2	0	0	0
x_6	0	0	4	2	-8	-1	1	0	0
x_7	0	0	11	5	-18	-2	0	1	1

↓

	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	1	0	0	-14.5	98	6.75	13.25	0	0
x_1	0	1	0	0.5	-4	-0.75	2.75	0	0
x_2	0	0	1	0.5	-2	-0.25	0.25	0	0
x_7	0	0	0	-0.5	4	0.75	-3.75	1	1

↓

	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	1	29	0	0	-18	-15	93	0	0
x_3	0	2	0	1	-8	-1.5	5.5	0	0
x_2	0	-1	1	0	2	0.5	-2.5	0	0
x_7	0	1	0	0	0	0	0	1	1

↓

	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	1	20	9	0	0	-10.5	70.5	0	0
x_3	0	-2	4	1	0	0.5	-4.5	0	0
x_4	0	-0.5	0.5	0	1	0.25	-1.25	0	0
x_7	0	1	0	0	0	0	0	1	1

↓

	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	1	-22	93	21	0	0	-24	0	0
x_5	0	-4	8	2	0	1	-9	0	0
x_4	0	0.5	-1.5	-0.5	1	0	1	0	0
x_7	0	1	0	0	0	0	0	1	1

↓

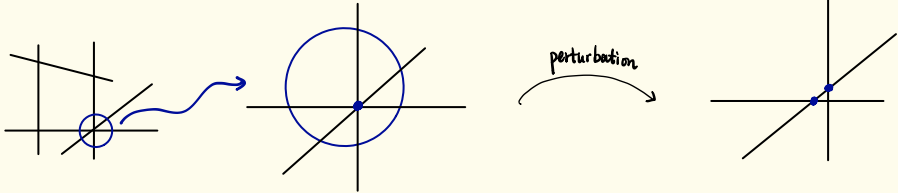
	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	1	-10	57	9	24	0	0	0	0
x_5	0	0.5	-5.5	-2.5	9	1	0	0	0
x_6	0	0.5	-1.5	-0.5	1	0	1	0	0
x_7	0	1	0	0	0	0	0	1	1

Note that the last tableau is identical to the first one.

- Anti-cycling rules

- Perturbation

Replace $\vec{b}$ with $\vec{b} + \begin{bmatrix} \varepsilon \\ \varepsilon^2 \\ \vdots \\ \varepsilon^m \end{bmatrix}$, where $\varepsilon > 0$ provides a small perturbation. The degenerate problem becomes nondegenerate.



- Bland's rule

Choose eligible (entering and departing) variable with the smallest index.

e.g.:

	z	x_1	x_2	x_3	x_4	RHS
z	1	-1	-2	0	0	5
x_4	0	1	3	0	1	2
x_3	0	2	-2	1	0	4

- Random rule

Choose eligible (entering and departing) variable at random.